

Generative AI's Act Two

From Technology Hammer to Real Value

Sequoia Capital | September 2023 | Sonya Huang, Pat Grady & GPT-4

Six Decades of Technological Progress Converged to Enable Generative AI

60 Years

Moore's Law

Delivered the compute horsepower to process exaflops of data — the raw engine powering modern AI training and inference.

40 Years

The Internet

Produced trillions of tokens of training data, accelerated by COVID-era digital adoption.

20 Years

Mobile & Cloud

Put a supercomputer in every human's palm, creating both distribution and edge compute.

ChatGPT's launch ignited a Cambrian explosion of innovation not seen since the early internet — but cracks soon appeared.

GenAI Startups Surpassed \$1B Revenue Faster Than SaaS Ever Did

>\$1B

in startup revenue alone — a milestone that took SaaS years, not months, to reach.

ChatGPT

Fastest-growing application in history

Strong product-market fit among students and developers.

Midjourney

Hundreds of millions in revenue

Achieved with a team of just 11 people — extraordinary capital efficiency.

Character.ai

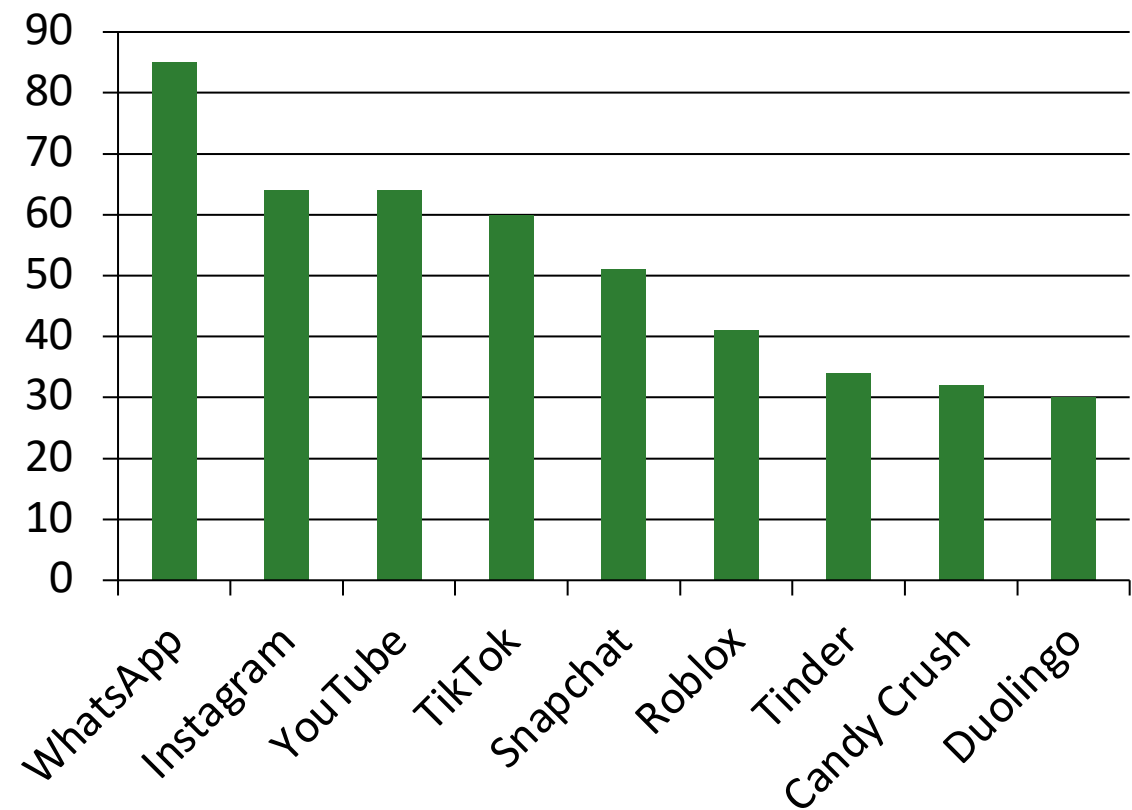
2 hours average session time

Popularized AI entertainment and companionship; users spend ~2 hours per session in-app.

Despite rapid revenue growth, many AI companies still lack product-market fit or sustainable competitive advantage.

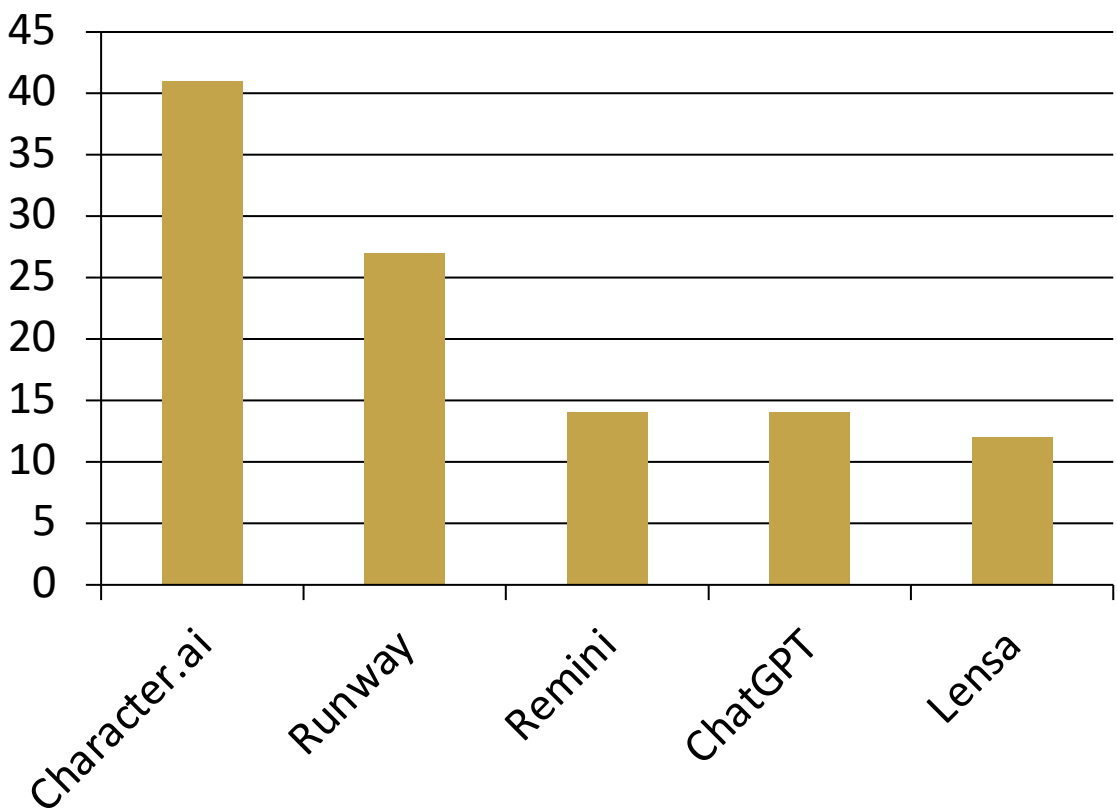
AI-First Apps Show 14% Median Retention vs. 51% for Incumbents

Incumbents



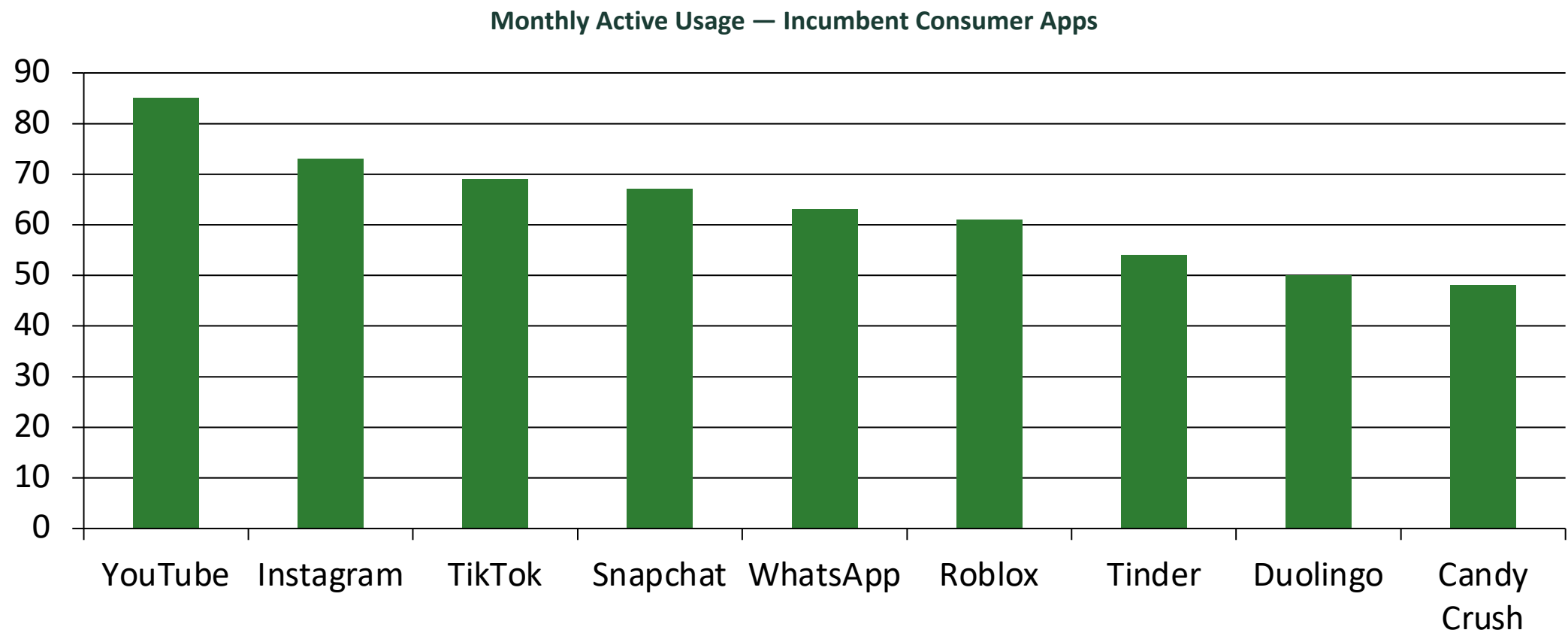
Median: 51%

AI-First Companies



Median: 14%

YouTube Leads Incumbents at 85% Monthly Active Usage



Median: 63%

This benchmark shows the high engagement bar AI-first apps must clear to compete with established consumer platforms.

Act 1 Was Technology-Out; Act 2 Is Customer-Back

ACT 1 — Technology-Out

- **Foundation models as the entire solution**

Apps were thin wrappers around a single model API with no differentiation.

- **Novelty demos & lightweight wrappers**

Products showcased cool technology but solved no enduring workflow problem.

- **Poor retention & no moat**

Users tried once and left; switching costs were near zero.

ACT 2 — Customer-Back

- **Foundation models as a piece of the solution**

Models are components within comprehensive, workflow-integrated products.

- **New editing interfaces & multi-modal workflows**

Interfaces that make outputs better and workflows stickier over time.

- **End-to-end problem solving**

Solving human problems holistically rather than offering thin wrappers.

Act 2 Exemplars

Harvey (custom LLMs for elite law firms) • Glean (crawling & indexing workspaces for enterprise AI) • Character & Ava (digital companions)

Market Map Reorganized by Use Case, Not Model Modality

OLD: Model-Modality Organization

- Text models → chatbots, copywriting
- Image models → art, design, avatars
- Code models → copilots, IDEs
- Audio models → voice, music



NEW: Use-Case Organization

- Legal → Harvey (custom LLMs for law firms)
- Enterprise Search → Glean (workspace AI)
- Companionship → Character & Ava (digital companions)
- Creative → Runway, Midjourney (end-to-end workflows)

This reorganization mirrors the broader Act 1 → Act 2 transition: the market map is now centered on the human problem being solved, not the underlying model type. It reflects the increasingly multi-modal nature of applications and the shift from technology hammer to actual value delivery.

Key Takeaways: What Separates Winners in Act Two

01

Retention is the critical challenge

AI-first apps sit at 14% median retention vs. 51% for incumbents — a 3.6× gap that must close for sustainable businesses.

02

Foundation models should be a component, not the whole product

Winning companies use models as pieces of comprehensive solutions rather than standalone demos.

03

Multi-modal, workflow-integrated solutions will win

New editing interfaces and end-to-end workflows create stickiness that thin wrappers cannot match.

04

The \$1B+ revenue milestone proves real demand exists

Despite hype-cycle skepticism, generative AI has demonstrated commercial traction faster than SaaS.

05

Solve end-to-end human problems to define Act 2

Companies like Harvey, Glean, Character, and Ava show that customer-back problem solving is the winning formula.